

DEFORESTATION

Deforestation is the clearing of forests or trees at a rate that exceeds the natural regeneration of forests.

It is a significant environmental issue that has a range of impacts on ecosystems, climate, and human communities. Deforestation occurs for a variety of reasons, including agricultural expansion, logging, mining and urbanization. Here are some of the impacts of deforestation.

1. BIODIVERSITY LOSS - Deforestation can result in the loss of plant and animal species, as well as the fragmentation of habitats, which can lead to declines in biodiversity.

2. CLIMATE CHANGE - Forests act as carbon sinks absorbing carbon dioxide from the atmosphere through ~~the~~ photosynthesis. Deforestation increases carbon dioxide in the atmosphere contributing to climate change.

3. SOIL EROSION - Deforestation can result in soil erosion and degradation which can lead to reduced soil fertility and agricultural productivity.

4. WATER CYCLE DISRUPTION - Trees play a critical role in regulating the water cycle and deforestation can disrupt this balance of water in ecosystems, leading to reduced water availability, increased flooding and changes in river flows.

5. IMPACTS ON LOCAL COMMUNITIES - Deforestation can have significant impacts on local communities including displacement, loss of livelihoods and impacts on cultural and spiritual practices.

TIMBER EXTRACTION - "Timber extraction refers to the process of removing trees from forests for commercial purposes, such as to produce wood products or to clear land for agriculture or other uses."

Timber extraction can have significant impact on forest ecosystems such as biodiversity loss, soil erosion, water cycle disruption and fragmentation.

To address the impacts of timber extraction, there are a range of conservation strategies that can be implemented including

1. SUSTAINABLE FORESTRY PRACTICES - Implementing sustainable forestry practices, such as selective logging and reforestation, can help to maintain the ecological integrity of forests.

2. COMMUNITY FORESTRY - Engaging local communities in forest management, such as through community forestry programs, can help to ensure that their needs and interests are taken into account.

3. LEGAL FRAMEWORK - Establishing and enforcing laws and regulations to govern timber extraction can help to ensure that it is carried out in a responsible and sustainable manner.

WATER RESOURCES

Water is a prerequisite for the existence of life. Human beings, animals and plants cannot survive without water. In fact, the human body is mostly water. Water is the most critical limiting factor for many aspects of life, such as economic growth, environmental stability, biodiversity conservation, food security and health care. In most cases, there is no substitute for water. An energy source can be replaced by another, but water as a resource is largely irreplaceable.

"Water resources are natural sources of water that are potentially useful for humans. Uses of water include agricultural, industrial, household, recreational and environmental activities."

SOURCES OF FRESHWATER

Fresh water is a renewable resource, yet the world's supply of clean freshwater is steadily decreasing. Water demand already exceeds supply in many parts of the world and as the world population continues to rise, so too does the water demand.

SURFACE WATER

Surface water is a ~~ground~~ water in a river, lake or fresh water wetland. Surface water is naturally replenished by precipitation and naturally lost through discharge to the ~~ocean~~ ~~ocean~~ ocean, evaporation and sub-surface seepage.

Although the only natural input to any surface water system, is precipitation within its watershed.

GROUNDWATER

Groundwater, is fresh water located in the pore space of soil and rocks. It is also water that is flowing within aquifers below the water table.

A layer of sediment or rock that is highly permeable and contains water is called aquifers.

Sand and gravel layers are good aquifers, while clay and crystalline rocks because of low permeability, are not good aquifers.

Aquifers may be unconfined or confined.

Unconfined aquifers are overlaid by permeable earth materials and are recharged by water seeping down from above in the form of precipitation.

Confined aquifers are sandwiched between two impermeable layers of rocks or sediments and are recharged only in those areas where the aquifer intersects the land surface.

Groundwater is one of the largest freshwater resources next to surface water, for agricultural and domestic use in many areas of the world.

USES OF WATER

There are many uses of water and it can be broadly divided into domestic, agricultural and industrial uses. In our country, a major portion of water is used for agriculture as India is an agricultural country.

DOMESTIC

The most important use of water is obviously in our homes. Domestic water ~~use~~ can be used for both indoor and outdoor household purposes. The daily uses include drinking, cooking, bathing, washing etc.

Irrigation

These include water allocations to farms, horticultural crops, pastures, and harvesting.

INDUSTRIAL USE

The uses include processing, cleaning, transportation, dilution and cooling. Major water using industries are paper, steel, chemical, and petroleum refining.

POWER GENERATION

The water has enormous potential to generate electric power with heat and water called hydroelectric power and geothermal energy.

RECREATION USE

Water plays an important role in recreational activities such as swimming, rafting, boating and so on.

It is understood that water is the most important natural resource gifted to us by nature. Clean water is scarce and has become one of the major problems in India. Let's save water and keep our water bodies clean as much as possible.

OVER UTILIZATION OF SURFACE AND GROUND WATER

With the growth of human population there is an increasing need for larger amounts of water to fulfill a variety of basic needs. The overutilization of water occurs at various levels. However, overuse of water can lead to a range of negative impacts.

1. Drought

Over use of surface water can lead to reduced water availability during droughts which can have significant impacts on agriculture, industry and human populations.

2. Ecological Impacts

Over use of surface water can also lead to significant ecological impacts such as the drying up of wetlands, damage to aquatic ecosystems and loss of habitat for wildlife.

3. Water Pollution

Overuse of surface water can increase water pollution, as pollutants

can become concentrated in the remaining water.

4. CONFLICTS Overuse of ~~surface~~ water can lead to conflicts between different users, such as farmers, industrial users and municipal water suppliers, who may all be competing for limited water resources.

5. Groundwater depletion

Excessive extraction of groundwater, especially in arid and semiarid regions, causes a sharp decline in the level of water table. * when the groundwater withdrawal rate exceeds its recharge rate, the sediments in the aquifer get compacted, a phenomenon known as "ground subsidence". It may cause huge economic losses because it results in the sinking of overlying land surface associated with structural damage to buildings.

* Over use of groundwater and freshwater along coast lines can lead to the intrusion of 'salt water' into aquifers, rendering the water unusable for human consumption or agriculture.

FLOODS

Floods occur when there is an overflow of water onto land that is normally dry. They can be caused by a range of factors, including heavy rainfall, snowmelt and storm surges. Floods can have a range of negative impacts including

1. **Property damage** - Floods can cause significant property damage including damage to homes and infrastructure.
2. **Displacement of people** - Floods can displace people from their homes and communities leading to social and economic disruption.
3. **Public health impacts** - Floods can spread the water-borne diseases and the contamination of drinking water sources.
4. **Ecological impacts** - Floods can have significant ecological impacts, such as the destruction of habitats, loss of biodiversity, and the spread of invasive species.